

# SCSD2613

## System Analysis and Design

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## PART IV

### The Analysis Process #2:

### Process Specifications & Structured Decisions

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# ■ OBJECTIVES

At the end of this topic, students shall be able to:

- i. Understand the purpose of process specifications in analyzing, describing and documenting structured decisions.
- ii. Produce process specification using structured English, decision tables, and decision trees
- iii. Choose an appropriate decision analysis method for analyzing structured decisions and creating process specifications.

# ■ LOGIC OF DECISION

- Methods of documenting and analyzing logic:
  - Structured English
  - Decision tables
  - Decision trees
- Important to distinguish between logic and structured decisions with semi-structured decisions
  - Semi-structured decisions may involve human judgement
- Structured decision analysis methods promote completeness, accuracy, and communication



wikiHow to Decide on Your Major

# ■ MAJOR TOPICS

## PROCESS SPECIFICATION

- Goals
- Process Specification Format

## STRUCTURED ENGLISH

- Writing Structured English
- Data Dictionary and Process Specification

## DECISION TABLE

- Developing Decision Table

## DECISION TREE

- Drawing Decision Tree

# PART 1

- Process Specification

# PROCESS SPECIFICATIONS

- Sometimes called **minispecs**
- Created for:
  - In DFD ☐ **primitive** processes and some higher-level processes on a data flow diagram
    - Primitive process - a process that is said to be functionally primitive, where it cannot be exploded further. Logic for this process need to be written to describe the primitive process.
  - In OO design ☐ class methods and for the steps in a use case.
- **Goals** of producing Process Specification:
  1. Reduce process ambiguity
  2. Obtain a precise description of what is accomplished
  3. Validate the system design

# ■ PROCESS SPECIFICATIONS

- However, it is **NOT** created for:



Processes that represent physical input and/or output

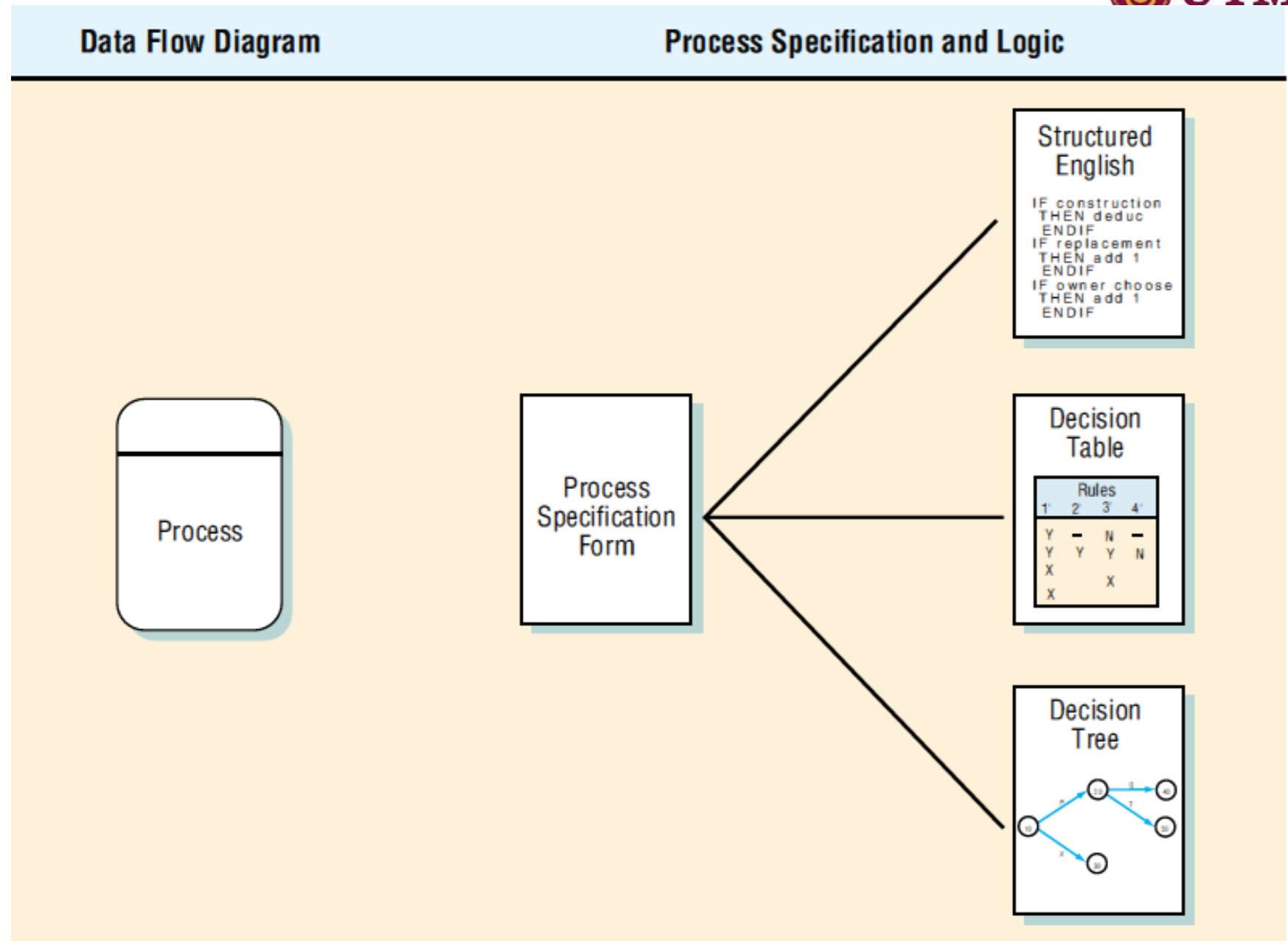


Processes that represent simple data validation



Processes that use prewritten code

# HOW PROCESS SPECIFICATION RELATE TO DFD





# PROCESS SPECIFICATION FORMAT

- Items on a process specification form:

|   | PROCESS SPECIFICATION ITEMS | DESCRIPTION                                                                                                                                                                                                                                                    |
|---|-----------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1 | The process number          | <ul style="list-style-type: none"> <li>• Must match the process ID on the data flow diagram</li> <li>• Allows the analyst to work on or review any process, and to locate the data flow diagram containing the process easily</li> </ul>                       |
| 2 | The process name            | <ul style="list-style-type: none"> <li>• Must be the same name as the process name on the DFD</li> </ul>                                                                                                                                                       |
| 3 | The process description     | <ul style="list-style-type: none"> <li>• Brief description of the process.</li> <li>• Example:<br/> <i>Determine if an item is available for sale. If it is not available, create a backordered item record. Determine the quantity available.</i> </li> </ul> |
| 4 | A list of input data flow   | <ul style="list-style-type: none"> <li>• Uses the names found on the data flow diagram</li> <li>• Data names used in the formula or logic should match the data dictionary, for consistency and good communication</li> </ul>                                  |
| 5 | Output data flows           | <ul style="list-style-type: none"> <li>• Uses the names found on the data flow diagram and data dictionary.</li> </ul>                                                                                                                                         |

|    | PROCESS SPECIFICATION ITEMS        | DESCRIPTION                                                                                                                                                                                                                                                                                                                       |
|----|------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 6  | Type of process                    | <ol style="list-style-type: none"><li>1. Batch</li><li>2. Online - require screen design or web pages</li><li>3. Manual - should have well-defined procedures for employees performing the process tasks</li></ol>                                                                                                                |
| 7  | Name of the subprogram or function | <ul style="list-style-type: none"><li>• If the process uses a subprogram or function, include the name of the subprogram or function containing the code</li></ul>                                                                                                                                                                |
| 8  | Process logic description          | <ul style="list-style-type: none"><li>• Description of the process logic that states the policy and business rules. Written in everyday language not computer language pseudo-code</li><li>• <b>Business rules</b>* are the procedures or a set of conditions or formulas, that allow a corporation to run its business</li></ul> |
| 9  | Logic method reference             | <ul style="list-style-type: none"><li>• If there is not enough room for a complete structured English description include a reference to the structured English description, decision table, or tree depicting the logic</li></ul>                                                                                                |
| 10 | List any unresolved issues         | <ul style="list-style-type: none"><li>• Incomplete portions of logic</li><li>• These issues form the basis of the questions used for follow-up interviews with users or business experts you have added to your project team</li></ul>                                                                                            |

\* please refer next slide

# ■ BUSINESS RULES

- What is BUSINESS RULES?
  - A business rule is a rule that defines a specific constraint within the context of a business.
- Common business rules format
  - Definitions of business terms
  - Business conditions and actions
  - Data integrity constraints
  - Mathematical and functional derivations
  - Logical inferences
  - Processing sequences
  - Relationships among facts about the business

# Example of Process Specification Form

| Process Specification Form                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |                          |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------|
| Number <u>1.3</u><br>Name <u>Determine Quantity Available</u><br>Description <u>Determine if an item is available for sale. If it is not available, create a backordered item record. Determine the quantity available.</u>                                                                                                                                                                                                                                                                                                                                                                                                                                                   |                          |
| Input Data Flow<br><u>Valid item from Process 1.2</u><br><u>Quantity on Hand from Item Record</u>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |                          |
| Output Data Flow<br><u>Available Item (Item Number + Quantity Sold) to Processes 1.4 &amp; 1.5</u><br><u>Backordered item to Inventory Control</u>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |                          |
| Type of Process<br><input checked="" type="checkbox"/> Online <input type="checkbox"/> Batch <input type="checkbox"/> Manual                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | Subprogram/Function Name |
| Process Logic:<br>IF the <u>Order Item Quantity</u> is greater than <u>Quantity on Hand</u><br>Then Move <u>Order Item Quantity</u> to <u>Available Item Quantity</u><br>Move <u>Order Item Number</u> to <u>Available Item Number</u><br>ELSE<br>Subtract <u>Quantity on Hand</u> from <u>Order Item Quantity</u><br>giving <u>Quantity Backordered</u><br>Move <u>Quantity Backordered</u> to <u>Backordered Item Record</u><br>Move <u>Item Number</u> to <u>Backordered Item Record</u><br>DO write <u>Backordered Record</u><br>Move <u>Quantity on Hand</u> to <u>Available Item Quantity</u><br>Move <u>Order Item Number</u> to <u>Available Item Number</u><br>ENDIF |                          |
| Refer to: Name: _____<br><input type="checkbox"/> Structured English <input type="checkbox"/> Decision Table <input type="checkbox"/> Decision Tree                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |                          |
| Unresolved Issues: <u>Should the amount that is on order for this item be taken into account? Would this, combined with the expected arrival date of goods on order, change how the quantity available is calculated?</u>                                                                                                                                                                                                                                                                                                                                                                                                                                                     |                          |

# PART 2

- Structured English

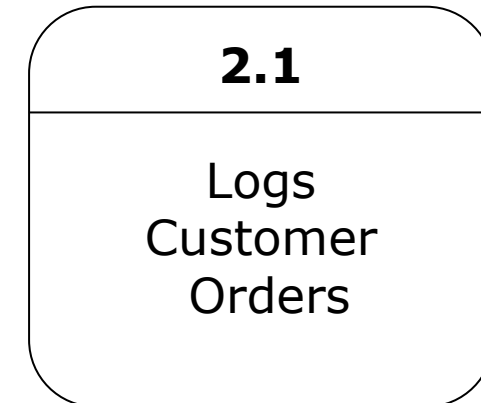
# ■ STRUCTURED ENGLISH

- Used when the process logic involves **formulas or iteration**, or when structured decisions are **not complex**
- Based on structured logic and simple English statements such as add, multiply, and move
- Advantages:
  1. Clarifying the logic and relationships found in human languages
  2. An effective communication tool, it can be taught to and understood by users in the organization

# ■ STRUCTURED ENGLISH

- It's similar to algorithm.
- It used to describe the logic process which involves formula, repetition, or simple structured decision.

```
DO
READ product's code
BEGIN IF
IF code start with "F"
    ADD name, quantity, product code
    ACCUMULATE the numbers of MEALS
IF code start with "D"
    ADD name, quantity, product code
    ACCUMULATE the numbers of DRINKS
IF code start with "DE"
    ADD name, quantity, product code
    ACCUMULATE the numbers of DESSERT
ENF IF
UNTIL EOF
THEN
    GENERATE Goods sold records
```





# ■ WRITING STRUCTURED ENGLISH

- Express all logic in terms of sequential structures, decision structures, case structures, or iterations
- Use and capitalize accepted keywords such as IF, THEN, ELSE, DO, and PERFORM
- Indent blocks of statements to show their hierarchy (nesting) clearly
- Underline words or phrases that have been defined in a data dictionary
- Clarify the logical statements



# ■ STRUCTURED ENGLISH EXAMPLES

| Structured English Type                                                                                                             | Example                                                                                                                                                                                   |
|-------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Sequential Structure<br>A block of instructions in which no branching occurs                                                        | Action #1<br>Action #2<br>Action #3                                                                                                                                                       |
| Decision Structure<br>Only IF a condition is true, complete the following statements; otherwise, jump to the ELSE                   | IF Condition A is True<br>THEN implement Action A<br>ELSE implement Action B<br>ENDIF                                                                                                     |
| Case Structure<br>A special type of decision structure in which the cases are mutually exclusive (if one occurs, the others cannot) | IF Case #1 implement Action #1<br>ELSE IF Case #2<br>Implement Action #2<br>ELSE IF Case #3<br>Implement Action #3<br>ELSE IF Case #4<br>Implement Action #4<br>ELSE print error<br>ENDIF |
| Iteration<br>Blocks of statements that are repeated until done                                                                      | DO WHILE there are customers.<br>Action #1<br>ENDDO                                                                                                                                       |

# ■ DATA DICTIONARY AND PROCESS SPECIFICATIONS

- The data dictionary is a starting point for creating structured English:
  - Sequence—a simple sequence of statements MOVE, ADD, and SUBTRACT
  - Selection—[] entries become IF...THEN...ELSE statements
  - Iteration { } entries become DO WHILE, DO UNTIL, or PERFORM UNTIL



# LET'S TRY

## Checkpoint 1: Structured English

Encik Amir is reviewing his firm's expense reimbursement policies with a new salesperson in his department.

*"Our reimbursement policies depend on the situation. You see, first we determine if it is a local trip. If it is, we only pay mileage of 18.5 cents a mile. If the trip was a one-day trip, we pay mileage and then check the times of departure and return. To be reimbursed for breakfast, you must leave by 7:00 A.M., lunch by 11:00 A.M., and have dinner by 5:00 P.M. To receive reimbursement for breakfast, you must return later than 10:00 A.M., lunch later than 2:00 P.M., and have dinner by 7:00 P.M. On a trip lasting more than one day, we allow hotel, taxi, and airfare, as well as meal allowances. The same times apply for meal expenses."*

Write structured English for Encik Amir's narrative of the reimbursement policies.

# PART 3

- Decision Table

# ■ DECISION TABLE

- A table of rows and columns, separated into four quadrants:
  - Conditions
  - Condition alternatives
  - Actions to be taken
  - Rules for executing the actions
- Advantages:
  - Help the analysis ensure completeness
  - Easy to check for possible errors
    - Impossible situations
    - Contradictions
    - Redundancy

| Conditions and Actions | Rules                  |
|------------------------|------------------------|
| Conditions             | Condition Alternatives |
| Actions                | Action Entries         |

Standard Format Used for Presenting a Decision Table

# ■ DEVELOPING DECISION TABLE

1. Determine conditions that affect the decision
2. Determine possible actions that can be taken
3. Determine condition alternatives for each condition
4. Calculate the maximum number of columns in the decision table
5. Fill in the condition alternatives
6. Complete table by inserting an X where rules suggest actions
7. Combine rules where it is apparent
8. Check for impossible situations
9. Rearrange to make more understandable

# ■ DECISION TABLE EXAMPLE 1

Banner's restaurant has two categories of employees . First, an employee who will be paid based on monthly salary (S). Second, who based on hours worked (H). There are three types of hours worked, less than 40, exactly 40 and more than 40. If (S) employees who work for 40 hours or less than 40 hours or more than 40 hours, they will be paid on monthly-based. If (H) employees and work less than 40 hours, the system will calculate hourly wage and an absence report must be produced. If (H) employees who has worked exactly 40 hours, the system will pay hourly wage. If (H) employees and work more than 40 hours, the system will hourly calculate wage and also calculate for overtime.

# ■ DECISION TABLE EXAMPLE 1

## Initial Table

| Condition              | Rules |     |    |    |     |     |
|------------------------|-------|-----|----|----|-----|-----|
|                        | 1     | 2   | 3  | 4  | 5   | 6   |
| Employee type          | S     | H   | S  | H  | S   | H   |
| Hours worked           | <40   | <40 | 40 | 40 | >40 | >40 |
| <b>Action</b>          |       |     |    |    |     |     |
| Pay base salary        | X     |     | X  |    | X   |     |
| Calculate hourly wage  |       | X   |    | X  |     | X   |
| Calculate overtime     |       |     |    |    |     | X   |
| Produce absence report |       | X   |    |    |     |     |



# ■ DECISION TABLE EXAMPLE 1

## Simplified Table

| Condition              | Rules |     |    |     |
|------------------------|-------|-----|----|-----|
|                        | 1     | 2   | 3  | 4   |
| Employee type          | S     | H   | H  | H   |
| Hours worked           | -     | <40 | 40 | >40 |
| <b>Action</b>          |       |     |    |     |
| Pay base salary        | X     |     |    |     |
| Calculate hourly wage  |       | X   | X  | X   |
| Calculate overtime     |       |     |    | X   |
| Produce absence report |       | X   |    |     |

# ■ DECISION TABLE EXAMPLE 2

Constructing a Decision Table for Deciding Which Catalog to Send to Customers Who Order Only from Selected Catalogs

| Conditions and Actions                   | Rules |   |   |   |   |   |   |   |
|------------------------------------------|-------|---|---|---|---|---|---|---|
|                                          | 1     | 2 | 3 | 4 | 5 | 6 | 7 | 8 |
| Customer ordered from Fall catalog.      | Y     | Y | Y | Y | N | N | N | N |
| Customer ordered from Christmas catalog. | Y     | Y | N | N | Y | Y | N | N |
| Customer ordered from specialty catalog. | Y     | N | Y | N | Y | N | Y | N |
| Send out this year's Christmas catalog.  |       | X |   | X |   | X |   | X |
| Send out specialty catalog.              |       |   | X |   |   |   | X |   |
| Send out both catalogs.                  | X     |   |   |   | X |   |   |   |

# DECISION TABLE EXAMPLE 2

Constructing a Decision Table for Deciding Which Catalog to Send to Customers Who Order Only from Selected Catalogs

| Conditions and Actions                   | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 |
|------------------------------------------|---|---|---|---|---|---|---|---|
| Customer ordered from Fall catalog.      | Y | Y | Y | Y | N | N | N | N |
| Customer ordered from Christmas catalog. | Y | Y | N | N | Y | Y | N | N |
| Customer ordered from specialty catalog. | Y | N | Y | N | Y | N | Y | N |
| Send out this year's Christmas catalog.  |   | X |   | X |   | X |   | X |
| Send out specialty catalog.              |   |   | X |   |   |   | X |   |
| Send out both catalogs.                  | X |   |   |   | X |   |   |   |

| Conditions and Actions                   | 1' | 2' | 3' |
|------------------------------------------|----|----|----|
| Customer ordered from Fall catalog.      | —  | —  | —  |
| Customer ordered from Christmas catalog. | Y  | —  | N  |
| Customer ordered from specialty catalog. | Y  | N  | Y  |
| Send out this year's Christmas catalog.  |    | X  |    |
| Send out specialty catalog.              | X  |    | X  |
| Send out both catalogs.                  |    |    |    |

# DECISION TABLE EXAMPLE 2

| Conditions and Actions                  | Rules |   |   |   |   |   |   |   |
|-----------------------------------------|-------|---|---|---|---|---|---|---|
|                                         | 1     | 2 | 3 | 4 | 5 | 6 | 7 | 8 |
| Customer ordered from Fall catalog      | Y     | Y | Y | Y | N | N | N | N |
| Customer ordered from Christmas catalog | Y     | Y | N | N | Y | Y | N | N |
| Customer ordered from Special catalog   | Y     | N | Y | N | Y | N | Y | N |
| Send out this year's Christmas catalog  |       | X |   | X |   | X |   | X |
| Send out Special catalog                |       |   | X |   |   |   | X |   |
| Send out both catalogs                  | X     |   |   |   | X |   |   |   |



Constructing a decision table for deciding which catalog to send to customers who order only from selected catalogs

| Conditions and Actions                  | Rules |   |   |   |   |   |   |   |
|-----------------------------------------|-------|---|---|---|---|---|---|---|
|                                         | 1     | 2 | 3 | 4 | 5 | 6 | 7 | 8 |
| Customer ordered from Fall catalog      | Y     | Y | Y | Y | N | N | N | N |
| Customer ordered from Christmas catalog | Y     | Y | N | N | Y | Y | N | N |
| Customer ordered from specially catalog | Y     | N | Y | N | Y | N | Y | N |
| Send out this year's Christmas catalog  |       | X |   | X |   | X |   | X |
| Send out Specialty catalog              |       |   | X |   |   |   | X |   |
| Send out both catalogs                  | X     |   |   |   | X |   |   |   |

| Conditions and Actions                  | Rules |    |    |
|-----------------------------------------|-------|----|----|
|                                         | 1'    | 2' | 3' |
| Customer ordered from Fall catalog      | -     | -  | -  |
| Customer ordered from Christmas catalog | Y     | -  | N  |
| Customer ordered from Specialty catalog | Y     | N  | Y  |
| Send out this year's Christmas catalog  |       | X  |    |
| Send out Specialty catalog              |       |    | X  |
| Send out both catalogs                  | X     |    |    |



Combining rules to simplify the customer-catalog decision table

# CHECKING FOR COMPLETENESS & ACCURACY

## 1. Impossible Situations

| Conditions and Actions | Rules |   |   |   |
|------------------------|-------|---|---|---|
|                        | 1     | 2 | 3 | 4 |
| Salary > \$50,000/year | Y     | Y | N | N |
| Salary < \$2,000/month | Y     | N | Y | N |
| Action 1               |       |   |   |   |
| Action 2               |       |   |   |   |

This is an impossible situation.

A person cannot earn greater than \$50,000 per year and less than \$2000 per month

# CHECKING FOR COMPLETENESS & ACCURACY

## 2. Contradiction and redundancy

| Conditions and Actions | Rules |   |   |   |   |   |   |
|------------------------|-------|---|---|---|---|---|---|
|                        | 1     | 2 | 3 | 4 | 5 | 6 | 7 |
| Condition 1            | Y     | Y | Y | Y | Y | N | N |
| Condition 2            | Y     | Y | Y | N | N | Y | N |
| Condition 3            | —     | N | — | — | — | N | Y |
| Action 1               | X     |   |   | X | X |   |   |
| Action 2               |       | X | X |   |   | X |   |
| Action 3               |       |   |   |   |   |   | X |

**Contradiction**

Have the same conditions but different results

**Redundancy**

Have the same conditions and results



# LET'S TRY

## Checkpoint 2: Decision Table

Using the same problem in the Checkpoint 1, create a decision table depicting the reimbursement for Encik Amir's office policies.

*(from Checkpoint 1)*

*"Our reimbursement policies depend on the situation. You see, first we determine if it is a local trip. If it is, we only pay mileage of 18.5 cents a mile. If the trip was a one-day trip, we pay mileage and then check the times of departure and return. To be reimbursed for breakfast, you must leave by 7:00 A.M., lunch by 11:00 A.M., and have dinner by 5:00 P.M. To receive reimbursement for breakfast, you must return later than 10:00 A.M., lunch later than 2:00 P.M., and have dinner by 7:00 P.M. On a trip lasting more than one day, we allow hotel, taxi, and airfare, as well as meal allowances. The same times apply for meal expenses."*

# PART 4

- Decision Tree

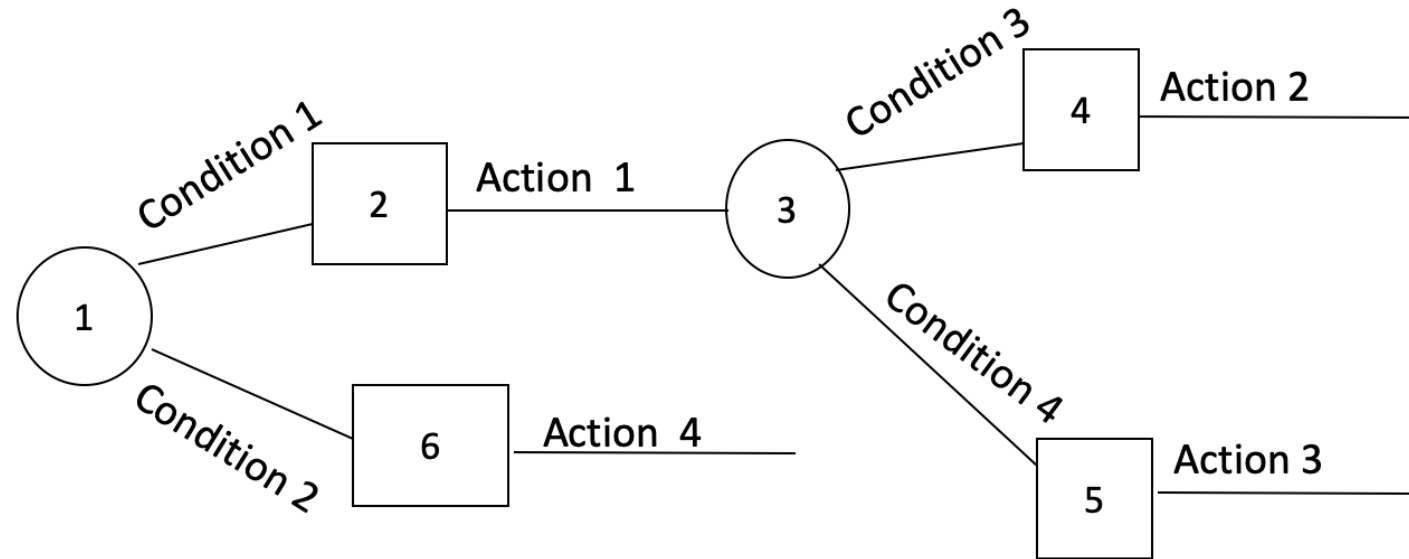


# ■ DECISION TREE

- Decision trees are used when complex branching occurs in a structured decision process
- Trees are also useful when it is essential to keep a string of decisions in a particular sequence
- Advantages:
  - The order of checking conditions and executing actions is immediately noticeable
  - Conditions and actions of decision trees are found on some branches but not on others
  - Compared to decision tables, decision trees are more readily understood by others in the organization

# ■ DRAWING DECISION TREE

- Identify all conditions and actions and their order and timing (if they are critical)
- Begin building the tree from left to right, making sure you list all possible alternatives before moving to the right



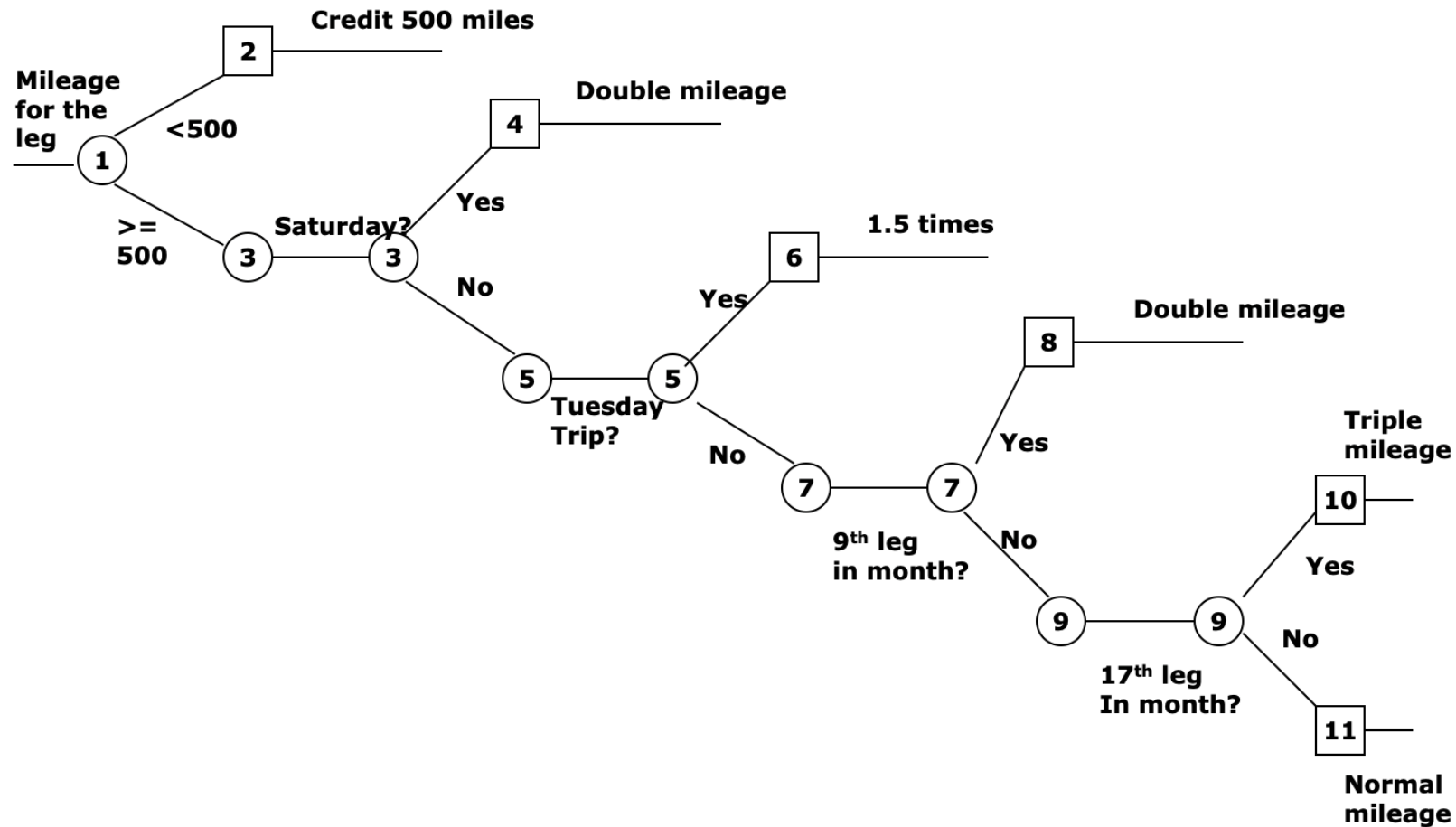
# ■ DECISION TREE EXAMPLE

**A policy for Premium Airlines : Accumulating Miles For Awards, as explained by Glen Curtis (marketing manager)**

“The traveler will be awarded the miles actually flown. If the actual mileage for the leg was less than 500 miles, the traveler will get 500 miles credit. If the leg was exactly or more than 500 miles and the trip was made on Saturday, the actual mileage will be multiplied by 2. If the trip was made on Tuesday, the multiplication factor is 1.5. If this is the ninth leg traveled during the calendar month, the mileage is doubled no matter what day, and if it is the seventeenth leg traveled, the mileage is tripled.”

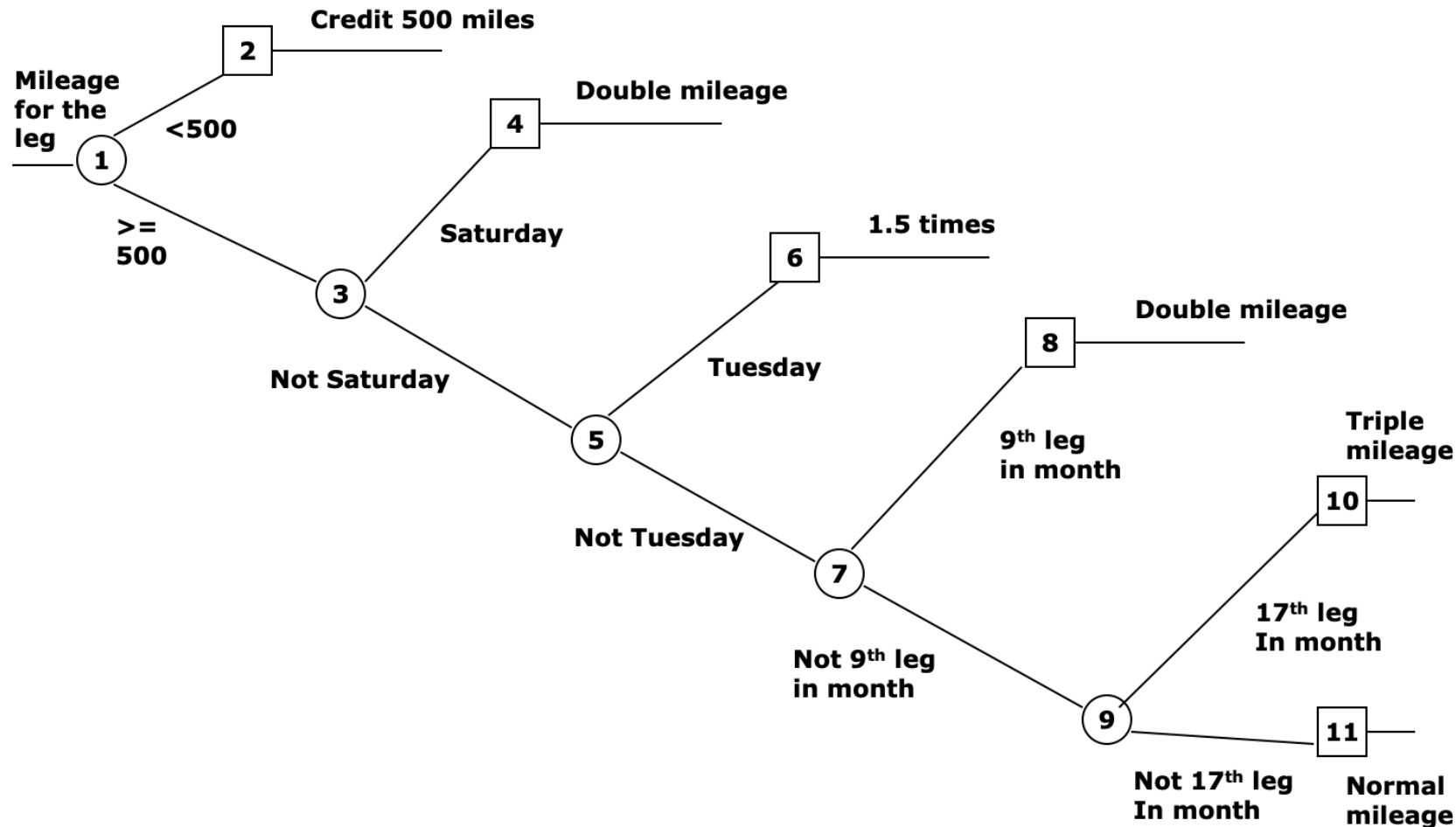
# DECISION TREE EXAMPLE

## Solution 1



# DECISION TREE EXAMPLE

## Solution 2





# LET'S TRY

## Checkpoint 3: Decision Tree

Based on your answer for the *Decision Table in Checkpoint 2*, draw a decision tree for Encik Amir's office reimbursement policies.

(from Checkpoint 1)

*"Our reimbursement policies depend on the situation. You see, first we determine if it is a local trip. If it is, we only pay mileage of 18.5 cents a mile. If the trip was a one-day trip, we pay mileage and then check the times of departure and return. To be reimbursed for breakfast, you must leave by 7:00 A.M., lunch by 11:00 A.M., and have dinner by 5:00 P.M. To receive reimbursement for breakfast, you must return later than 10:00 A.M., lunch later than 2:00 P.M., and have dinner by 7:00 P.M. On a trip lasting more than one day, we allow hotel, taxi, and airfare, as well as meal allowances. The same times apply for meal expenses."*

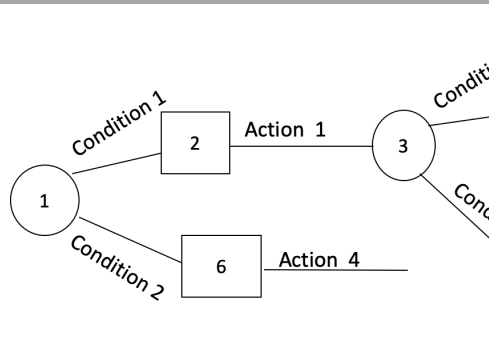
# ■ WHEN SHOULD BE USED?

```

DO
READ product's code
BEGIN IF
IF code start with "F"
    ADD name, quantity, prod
    ACCUMULATE the numbers
IF code start with "D"
    ADD name, quantity, prod
    ACCUMULATE the numbers o
IF code start with "DE"
  
```

## STRUCTURED ENGLISH

- It is well suited to modeling processes with many sequential steps and relatively simple control logic. It is not suitable for describing complex decision logic or few or no sequential processing steps. This technique is **complimentary with DFD** to designing a mini-spec for the certain process element of DFD.



## DECISION TREE & TABLE

- It is used to summarizing complex decision logic more concisely than structured English. It is easier to describing the “branch” logic by using decision tree rather than decision table. These techniques are **complimentary with DFD** when importantly used to designing a mini-spec for the certain process element of DFD.

# ■ SUMMARY – DECISION ANALYSIS

- Selecting a structured decision analysis technique – What to consider?

|                    | WHEN TO USE?                                                                                                                                                                   | ADVANTAGES                                                                                                                                                |
|--------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------|
| STRUCTURED ENGLISH | When there are many repetitious actions or when communication to end users is important                                                                                        | Useful when many actions are repeated and when communicating with others is important                                                                     |
| DECISION TABLES    | When a complex combination of conditions, actions, and rules are found or you require a method that effectively avoids impossible situations, redundancies, and contradictions | Provide complete analysis of complex situations while limiting the need for change attributable to impossible situations, redundancies, or contradictions |
| DECISION TREES     | When the sequence of conditions and actions is critical or when not every condition is relevant to every action (the branches are different)                                   | Important when proper sequencing of conditions and actions is critical and when each condition is not relevant to each action                             |





# HOMEWORK

## Checkpoint 4: Final Exam 2016/2017

Your company has been hired as IT consultant to assist 'Exotic Treat' company in developing a newly system that allows their customer to purchase homemade sweets and cakes through online booking application. One of your job as the in-charge Business Analyst is to assist the development team in analyzing the detail process and logic flows for the determination of total payment invoice. The criteria in calculating the total payment invoice are as follows:

- Customers with e-coupon will be granted for \$10 discounts less for each total payment invoice
- Customers who spent more than \$250 purchase items with:
  - i. Numbers of order items (more OR less than 4 order items);
  - ii. Choices of delivery day once the order is placed (next day, second day, OR seventh day)

The specific conditions to calculate total payment invoice are summarized using the decision table in **Table 1**.

- Construct an optimized decision tree to demonstrate the logic conditions and actions sequences of process flows in calculating the total payment invoice (8 marks)
- In your opinion, which method (decision table or decision tree) is more helpful to analyze the process flows in calculating total payment invoice? Justify your reason. (2 marks)

| Total \$purchase > \$250 | YES                                                          |                 |                 |       |                 |                 | NO                                   |                 |                 |       |                 |                 |
|--------------------------|--------------------------------------------------------------|-----------------|-----------------|-------|-----------------|-----------------|--------------------------------------|-----------------|-----------------|-------|-----------------|-----------------|
| Number of items (n)      | N ≤ 3                                                        |                 |                 | N ≥ 4 |                 |                 | N ≤ 3                                |                 |                 | N ≥ 4 |                 |                 |
| Delivery day             | Next                                                         | 2 <sup>nd</sup> | 7 <sup>th</sup> | Next  | 2 <sup>nd</sup> | 7 <sup>th</sup> | Next                                 | 2 <sup>nd</sup> | 7 <sup>th</sup> | Next  | 2 <sup>nd</sup> | 7 <sup>th</sup> |
| Delivery \$charge        | 25                                                           | 10              | n*1.5           | n*6.0 | n*2.5           | Free            | 35                                   | 15              | 10              | n*7.5 | n*3.5           | n*2.5           |
| e-Coupon                 | YES                                                          |                 |                 |       |                 |                 | NO                                   |                 |                 |       |                 |                 |
| Total invoice (\$)       | (Total \$purchase + Delivery \$charge) – \$10 discounts less |                 |                 |       |                 |                 | Total \$purchase + Delivery \$charge |                 |                 |       |                 |                 |

**Table 1:** Decision table for calculating shipping charges and total invoice



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# Thank You

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